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sfcount: Command for Count Data Stochastic Frontiers and Underreported and Overreported Counts.

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Abstract. We introduce a new command, `sfcount`, to estimate Count Data Stochastic Frontier models. Although originally designed to estimate production and production-cost functions, this new command can be used to estimate mean regression functions when count data are suspected to be under/overreported.

Keywords: st0001, sfcount, stochastic frontier, underreporting, overreporting, low discrepancy series, Maximum Simulated Likelihood.

1 Introduction.

In this article we introduce a new Stata command, `sfcount`, which estimates the parameters of the Count Data Stochastic Frontier (CDSF) model in Fé and Hofler (2013).

Stochastic frontier models (Aigner et al. 1977; Meeusen and van den Broeck 1977) are central to the identification of inefficiencies in the production (and production costs) of continuously distributed outputs. In labour, industrial and health economics, production frontiers have also been adopted to explain deviations from maximal or minimal levels of non-tangible and/or non-pecuniary outcomes. However in these latter domains outcomes are often measured as counts (for example, the number of patents obtained by a firm or the number of infant deaths in a region). Although these latter fields of inquiry have not emphasised the idea of inefficiency in the 'production' of non-tangible and/or non-pecuniary outcomes, recent contributions (e.g. Fé 2013; Fé and Hofler 2013) suggest that inefficiencies are also present in these domains.

The need for specific count data models for stochastic frontiers arises because, not only this results in more efficient estimation (e.g. Greene 2004), but, more critically, inefficiency is typically not nonparametrically identified from data alone. Therefore researchers have to make specific assumptions regarding the distribution of inefficiency in the sample or the population. These assumptions define the class of admissible distribution underlying outcomes. Standard continuous data models attribute any negative (positive) skewness in the sample to inefficiencies in the production of economic goods (bads). The distributions of discrete outcomes are, however, typically skewed even in the absence of inefficiencies (e.g. the Poisson distribution) and the sign of skewness is generally independent of whether one is studying an economic good or an economic bad.

 st0001

As a result, standard stochastic frontier models can fail to detect any inefficiency in production when the outcome of interest is a count -even when the underlying inefficiency is substantial.

The core count data stochastic frontier model is based on a mixed Poisson distribution with a log-halfnormal mixing parameter (or a Poisson log-halfnormal, PHN, in the parlance of Fé and Hofler 2013). Although the motivation behind this model was the estimation of stochastic frontiers under discrete valued outcomes, the PHN can be employed for modelling underreported counts as well as overreported counts. These situations are pervasive when studying worker's absenteeism (Winkelmann 1996), consumer data (Fader and Hardie 2000), drug abuse (Brookoff et al. 1993) or traffic accidents (Alsop and Langley 2001) among others. Among the models traditionally used for modelling these events the Beta-Binomial and Poisson-Lognormal models have been widely used. The PHN is a complement to these specifications.

The code presented here extends the catalogue of Stata commands pertaining to the stochastic frontier literature including the original Stata commands `frontier` and `xtfrontier` as the recent extensions `sfcross` and `sfpanel` by Belotti 2013). The original model in Fé and Hofler (2013) was cross-sectional in nature. Therefore, this code does not take into account individual time-invariant heterogeneity. In the continuous outcome stochastic frontier literature, panel data extensions abound. For an excellent review -including extensions- see Greene (2005); a recent important methodological contribution is Belotti and Ilardi (2018). Similarly, the original model in Fé and Hofler (2013) did not deal with endogenous regressors. This is an active area of research in the general stochastic frontier literature. Seminal contributions include Kutlu (2010), Griffiths and Hajargasht (2016) and Amsler et al. (2016)[1]. The development of a count data model with covariates endogenous to inefficiency is an unexplored area of work.

2 Methods

To introduce the PHN model we adopt the stochastic frontier terminology in Fé and Hofler (2013). The relationship to under/overreported counts models will be apparent from the context. There is a sample of $i = 1, \ldots, n$ units containing data on a discrete outcome of interest $y_i \in \{0, 1, 2, \ldots\}$. The mean production frontier of y is determined by the mapping

$$\log \tilde{\lambda} = h(x; \beta).$$

where $\tilde{\lambda} \in \mathbb{R}^+$. Conditional on a level of inefficiency (or level of under-/over- reporting) $\varepsilon \in \mathbb{R}^+$, the mean deterministic frontier is

$$\log \lambda = h(x; \beta) \pm \varepsilon$$

Since we are modelling non-negative count data, we transform the last equation to

$$\lambda = \exp(h(x; \beta) \pm \varepsilon).$$

1. Another recent contribution is Karakaplan and Kutlu 2017. For accompanying code see Karakaplan (2017)

Following convention, we assume that y, has an Poisson distribution conditional on a set of regressors, x, and ε, with λ as the conditional mean of the distribution. The unconditional distribution follows by endowing ε with an specific density. Following the convention in the Stochastic Frontier literature, Fé and Hofler (2013) assume that ε follows a half normal distribution, so that

$$f(\varepsilon) = f(\varepsilon;\sigma) = \frac{2}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right) \mathbb{I}_{[0,\infty)} \text{ for } \sigma_\varepsilon > 0. \quad (1)$$

This density has the advantage of allowing some flexibility, thanks to its scale parameter. It also leads to a model whose first order moments are well defined (a property that is not shared by some popular distributions[2]).

If $f(\varepsilon)$ is half normal we can write $\varepsilon = |u|$, where u has a normal distribution. With this notation, and letting $h(x;\beta) = x'\beta$, the conditional distribution of y given x follows by averaging $\mathbb{P}(y|x,u)$ over the range of u,

$$\begin{aligned} \mathbb{P}(y|x;\sigma,\beta) &= E\left[\frac{(\exp(-\exp(x'\beta \pm \sigma|u|)))\exp(y(x'\beta \pm \sigma|u|))}{y!}\right] \\ &= E\left[\text{Poisson}\left(\exp(x'\beta \pm \sigma|u|)\right)\right] \quad (2) \end{aligned}$$

where expectations are taken with respect to the standard normal distribution. Fé and Hofler 2013 provide expressions for the moments of $f(\varepsilon)$ when this follows a half normal density function, as well as expressions for the conditional mean and variance of y. The PHN distribution does not have a closed form expression, however the integral in (2) can be approximated by simulation. Specifically, Fé and Hofler (2013) advocate combining Maximum Simulated Likelihood estimation of the PHN model with Halton sequences (Gentle 2003). Applying simulation, the conditional distribution of y_i (for $i = 1, \ldots, n$) is approximated by the sum:

$$\mathbb{P}(y|x;\theta) \approx \hat{\mathbb{P}}(y|x;s_h,\theta) = \frac{1}{H}\sum_{h=1}^{H} \text{Poisson}(\exp(x'\beta \pm \sigma|s_h|)) \quad (3)$$

where s_h are the terms of a, possibly randomized, Halton sequence. The infeasible log-likelihood $L_n = \sum_{i=1}^n \log \mathbb{P}(y|x;\theta)$ can be approximated by $L_{n,h} = \sum_{i=1}^n \log \hat{\mathbb{P}}(y|x;s_h,\theta)$. The analytical derivatives of $L_{n,h}$ are given by

$$\frac{\partial L_{n,h}}{\partial \theta} = \sum_{i=1}^{n} \frac{1}{\hat{\mathbb{P}}(y_i|x_i;s_h,\theta;)} \frac{1}{H} \sum_{h=1}^{H} \text{Poisson}(\tilde{\lambda}_{i,h})(y_i - \tilde{\lambda}_{i,h}) \left\{ \begin{array}{c} x_i \\ \pm|s_h| \end{array} \right\} \quad (4)$$

where $\tilde{\lambda}_{i,h} = \exp(x_i'\beta \pm \sigma|s_h|)$. The value $\hat{\theta}_{MSL}$ making the above system of equations equal to zero is the Maximum Simulated Likelihood estimator of θ. When the PHN is a correct representation of the underlying data generating process, $\hat{\theta}_{MSL}$ is a consistent, asymptotically normal and efficient estimator of the true parameter value, as

2. For example, when that ε has gamma distribution with parameters $\alpha > 0$ and $\delta > 0$ such that $\delta = \alpha$; see Fé and Hofler 2013

follows from properties 1 and 2 above. Standard errors can be computed via the BHHH estimator or the minus inverse of the Hessian Matrix of $L_{n,h}$.

Tests of hypotheses can rely on the Wald-Score-LR trinity. Testing the null hypothesis of no inefficiency is of particular interest. The null hypothesis would be

$$H_0 : \sigma = 0$$

in which case PHN collapses to a standard Poisson model. A formal test can be computed via a likelihood ratio comparing the resulting value with the quantiles of a χ_1^2 distribution.

2.1 Estimating cross-sectional inefficiency.

Although the parameters of the frontier are of interest in themselves, the ultimate goal of most stochastic frontier analyses is to obtain approximate efficiency scores (i.e. measures of the deviations away from either maximal or minimal values of the outcome) for each individual in the sample. Following Jondrow et al. (1982) cross-sectional inefficiency scores, $v = \exp(\pm|u|)$, can be estimated via $E(v|y,x)$. Using Bayes' theorem

$$f(v|x,y) = \frac{\mathbb{P}(y|x,v)f(v)}{\mathbb{P}(y|x)} \tag{5}$$

so that

$$E(v|y,x) = \int v f(v|x,y) dv. \tag{6}$$

The latter expression does not have a closed form. However, we may still approximate the relevant integral via simulation. The simulated $E(v|y,x)$ for the parametric Mixed Poisson model is

$$\hat{v}_i \quad = \quad E(v_i|x_i,y_i) \approx \frac{\sum_{h=1}^{H} \exp(\pm|s_h|\sigma)\text{Poisson}(\exp(x_i'\beta + \sigma|s_h|))}{\sum_{h=1}^{H} \text{Poisson}(\exp(x_i'\beta + \sigma|s_n|))} \tag{7}$$

Two remarks are important here. First, the distributions of v and $\hat{v}$ are not the same[3] and the lower and upper tails of the distribution of v will be misreported. From a stochastic frontier perspective, this means that $\hat{v}$ penalizes outstanding firms and rewards the least efficient individuals -although the average efficiency in the sample is correctly approximated. However, the estimator is unbiased in the unconditional sense $E(\hat{v} - v) = 0$ (Wang and Schmidt 2009).

Second, in applications, the scores depend critically on the term Poisson($\exp(x_i'\beta + \sigma|s_h|)$). If the mean of this distribution, $\exp(x_i'\beta + \sigma|s_h|)$ is too large in relation to Y for any one observation, then Poisson($\exp(x_i'\beta + \sigma|s_h|)$ will be approximately 0. Therefore, for that observation the cross-sectional estimate of inefficiency will be 0/0 which Stata reports as missing value. We thus recommend researchers to ensure that the explanatory variables are measured in meaningful units albeit of small magnitude.

3. As can be seen by noting that $\text{var}(v) = \text{var}(E(v|x,y)) + E(\text{var}(v|x,y))$, (and hence, $\hat{v}$ has smaller variance)

3 Stata syntax

<u>`sfcount`</u> *depvar* [*indepvars*] [*if*] [*in*] [, <u>`draws`</u>`(`*#*`)` <u>`technique`</u>`(`*string*`)` <u>`cost`</u>
`cluster(`*vcetype*`)`]

where *depvar* is the dependent variable and *indepvars* are the explanatory variables.

3.1 Options.

<u>`draws`</u>`(`*#*`)` Specify the number of Halton draws (the default is 200). The model is estimated via Maximum Simulated Likelihood. To approximate the likelihood function of the Poisson-log-Half-Normal model, the command uses Halton sequences (a low-discrepancy sequence). Halton sequences ensure a good coverage of the unit interval (e.g. Niederreiter 1992).

<u>`technique`</u>`(`*string*`)` specifies the optimization technique. The default option is the modified Newton-Raphson (`nr`). You can switch among `dfp` (Davidon-Fletcher-Powell), `bhhh` (Berndt-Hall-Hall-Hausman) and `bfgs` (Broyden-Fletcher-Goldfarb-Shanno).

<u>`cost`</u> Specifies that the underlying model is a cost function (or, equivalently, an over-reporting or deviation above the minimal level function). By default `sfcount` estimates a production function (or equivalently, an underreporting or deviation below the maximal level function).

<u>`cluster`</u>`(`*string*`)` specifies the name of a variable that creates intra-group correlation, relaxing the usual requirement that the observations be independent.

3.2 Cross-sectional estimates of inefficiency.

The command automatically generates a variable named `inefficiency` collecting the cross sectional scores.

3.3 Example

For this example, we generated 1000 observations from a PHN distribution with mean $\exp(1 + x_1 + x_2 - v)$ where $x_j \sim \text{Uniform}[0, 1]$ and v has a halfnormal distribution with $\sigma = 1$ (the code to generate the data appears in the discussion of the Monte Carlo simulation below). This yielded the following output

```
sfcount dep x1 x2
                                                Number of obs     =      1,000

------------------------------------------------------------------------------
         dep |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
eq1          |
          x1 |   1.009668    .070637    14.29   0.000     .8712215    1.148114
```

x2	.9658712	.0731504	13.20	0.000	.822499	1.109243
_cons	1.024507	.0665457	15.40	0.000	.8940796	1.154934
eq2						
_cons	-.0217397	.051502	-0.42	0.673	-.1226818	.0792023

```
Note: _cons in eq2 corresponds to the log of the standard deviation
of the mixing log-half-normal parameter
Ho: Inefficiency not present in the sample
chi2(1) =        325.51
Prob > chi2 =      0.00
```

The output follows standard Stata convention. The first block of results, `eq1`, presents the estimates of the structural coefficients of the regressors (including an intercept). The second block of results, `eq2`, presents the estimate of $\log(\sigma)$. Therefore, the point estimate of σ can be retrieved by using the transformation $\sigma = \exp($`_cons`$)$ in `eq2`. Below the table of main results one finds the likelihood ratio test for $H_0 : \sigma = 0$. In this case, the statistic equals 325.51 with associated p-value of 0.00 (and thus one would reject the null hypothesis).

The `sfcount` command automatically calculates the cross-sectional inefficiency scores and stores those into a new variable, `inefficiency`. The following summary statistics compare the average and actual estimated inefficiencies.

Variable	Obs	Mean	Std. Dev.	Min	Max
expv	1,000	.5210644	.2519012	.0130767	.9992961
inefficiency	1,000	.5297692	.1732635	.1411413	.9017366

The variable `expv` collects the true inefficiency scores, whereas the second variable, `inefficiency`, collects the estimated inefficiency scores. It is clear that, as pointed out by Wang and Schmidt (2009), the cross-sectional estimator is unbiased in an unconditional sense and thus it provides very accurate estimates of the average inefficiency. Even though figure 1 reveals a discrepancy between the actual and the estimated scores (which is expected since, in this simulation, x_j do not provide any information about the inefficiency parameter) there is also a strong positive correlation between the actual and estimated inefficiencies. The smaller standard deviation in the estimated inefficiencies is due to $\hat{v}$ begin a shrinkage of v towards its mean (as noted by Wang and Schmidt 2009).

4 Monte Carlo Simulation.

To verify the properties of the method and the robustness of the software, we run a Monte Carlo simulation. We drew samples of 100 observations from a PHN model with conditional mean $\exp(1+x_1+x_2-\sigma v)$ where $x_j \sim \text{Uniform}[0,1]$ and v has a halfnormal distribution with $\sigma \in \{0.5, 1, 2\}$. We estimated the parameters of this model 500 times. We wrote a short ado file, `mc1`, to generate the data, estimate the model and return the

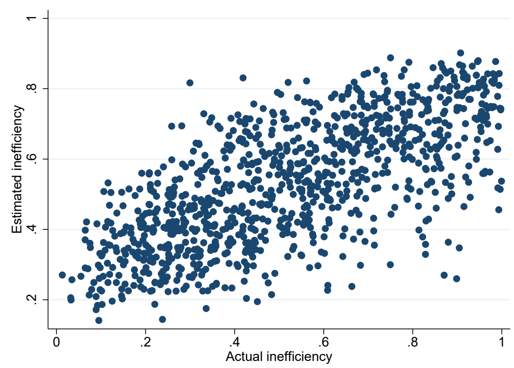

Figure 1: Real vs estimated cross-sectional inefficiency.

result to Stata's **simulate** command[4]. The results of each experiment are presented now. For $\sigma = 0.5$, we obtained the following results.

Variable	Obs	Mean	Std. Dev.	Min	Max
b0	500	.9968649	.1848761	.4428408	1.652993
b1	500	1.00012	.1827392	.4863433	1.519381
b2	500	.9707445	.1909647	.4019189	1.443396
s	500	.4656329	.1557467	.0001077	.8336025
reject	500	.654	.4761696	0	1

Here `b0` corresponds to the intercept, `b1, b2` are the coefficients of `x1,x2` respectively, `s` is the estimate of σ and `reject` is the proportion of times the null hypothesis $\sigma = 0$ was rejected by the likelihood ratio test. It is promptly observed that, even for the very small sample size considered, the parameters of the model are very accurately estimated on average. Specifically, the critical parameter σ was tightly concentrated around the true value of 0.5. The empirical power of the likelihood ratio test was 65% which is acceptable give the very small sample size considered. Similar conclusions were reached with the alternative specifications[5].

What would occur if the standard normal-halfnormal model in Aigner et al. (1977) were fitted to these data instead? We illustrate the situation through a simulation for both the cost and production frontier cases. We maintain the same design but we will focus on the case $\sigma = 1$ for simplicity. Following convention in Aigner et al. (1977), we fit the model

$$\log y = \theta_0 + \theta_1 \log x_1 + \theta_2 \log x_2 \pm u + v$$

where now, u is the inefficiency term (distributed halfnormal) and v is the idiosyncratic, zero mean error term. The case $+u$ corresponds to the cost function, whereas the case $-u$ corresponds to the production frontier. The code to run this simulation is similar to `mc1.ado`, however `sfcount` is replaced by the built-in command `frontier` and the

4. The code for the simulation can be obtained from the authors upon request.
5. The results can be also obtained from the authors upon request.

variables are transformed to logs. The critical parameters in this model are the standard errors of u and v, say σ_u, σ_v. Specifically, the ratio $\lambda = \sigma_u/\sigma_v$ is the commonly used measure of the magnitude of inefficiency in the sample. In addition to this, the likelihood ratio test of $\sigma_u = 0$ serves, as before, as the statistic to draw inferences regarding the statistical significance of inefficiency in the sample. The result of this is

Variable	Obs	Mean	Std. Dev.	Min	Max
b0	500	.2211184	.0890768	.0012678	.6122401
b1	500	.2313553	.0854484	.0486661	.6074979
b2	500	2.499497	.2308907	1.412019	3.118477
s	500	2158507	8511800	.0325458	5.28e+07
reject	500	.726	.4464556	0	1
k3	500	-.1886938	.1433905	-.6482261	.213362

The critical quantities in the simulation are `k3` and `s` (which corresponds to the ratio $\lambda = \sigma_u/\sigma_v$). The production stochastic frontier model expects negative skewness in the log of the dependent variable, and on this occasion, the average skewness happens to be negative (but this is not always, as illustrated below). Indeed, the type of skewness exhibited by $\log Y$ will depend on the specific parameterization of the generating process (note that the untransformed data will always have a positively skewed distribution). The most critical aspect of these results is that the parameter λ is estimated very imprecisely. The large value of the estimated parameter would suggest that too often the model cannot separate inefficiency from pure noise, and that all the variation in the sample is erroneously attributed to inefficiency. In contrast, the likelihood ratio test only rejects the null hypothesis about 70% of the time. The result for the cost frontier are similarly worrying,

Variable	Obs	Mean	Std. Dev.	Min	Max
b0	498	.2829813	.0885495	.0617461	.6670547
b1	498	.2877539	.0910561	.0603849	.6406236
b2	498	2.79976	.3151728	2.19097	3.677881
s	498	97321.05	1311344	.0013252	2.15e+07
reject	498	.3815261	.4862496	0	1
k3	498	.0985375	.2989368	-.7290995	.9630092

In this case, maximum likelihood failed to converge in 2 replications. As with the production frontier case, the parameter λ is imprecisely estimated (because all variation tends to be attributed to inefficiency). Yet, paradoxically, the significance of the inefficiency term is rejected only about 38% of the time. The critical aspect of this result is that although the distribution of $\log Y$ has the right type of skewness (positive), its magnitude is very small; therefore the cost stochastic frontier model fails to identify any inefficiency, even though this is prevalent in the data.

An even more problematic example arises when we let $\sigma = 3$ and try to estimate a production frontier. Here the skewness of $\log Y$ is positive, which presents a violation of one of the assumptions underlying the continuous data stochastic frontier model. In this instance, the latter method fails to detect any inefficiency at all,

Variable	Obs	Mean	Std. Dev.	Min	Max
b0	500	.1504993	.1210131	-.2304306	.8188418
b1	500	.1507706	.1221521	-.2453529	.7454497
b2	500	1.335596	.4623144	.6625592	3.285585
s	500	479674	3430413	.0114056	3.44e+07
reject	500	.056	.230152	0	1
k3	500	.2978432	.2033263	-.285708	.9867306

As seen from the above results, the likelihood ratio test rejects the null hypothesis at the nominal 5% level, whereas λ remains to be imprecisely estimated.

In summary, the continuous data stochastic frontier model can be problematic in practice when data are coming from the PHN specification.

5 Conclusion.

We have introduced a new command, `sfcount`, to estimate the Count Data Stochastic Frontier models in Fé and Hofler (2013). We have illustrated the implementation of this method and, through simulations, we have further illustrated the need for such a method. Although originally designed to estimate production and production-cost functions, our command can be used to estimate mean regression functions when count data are suspected to be under/overreported. The latter situations are common in empirical applications.

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